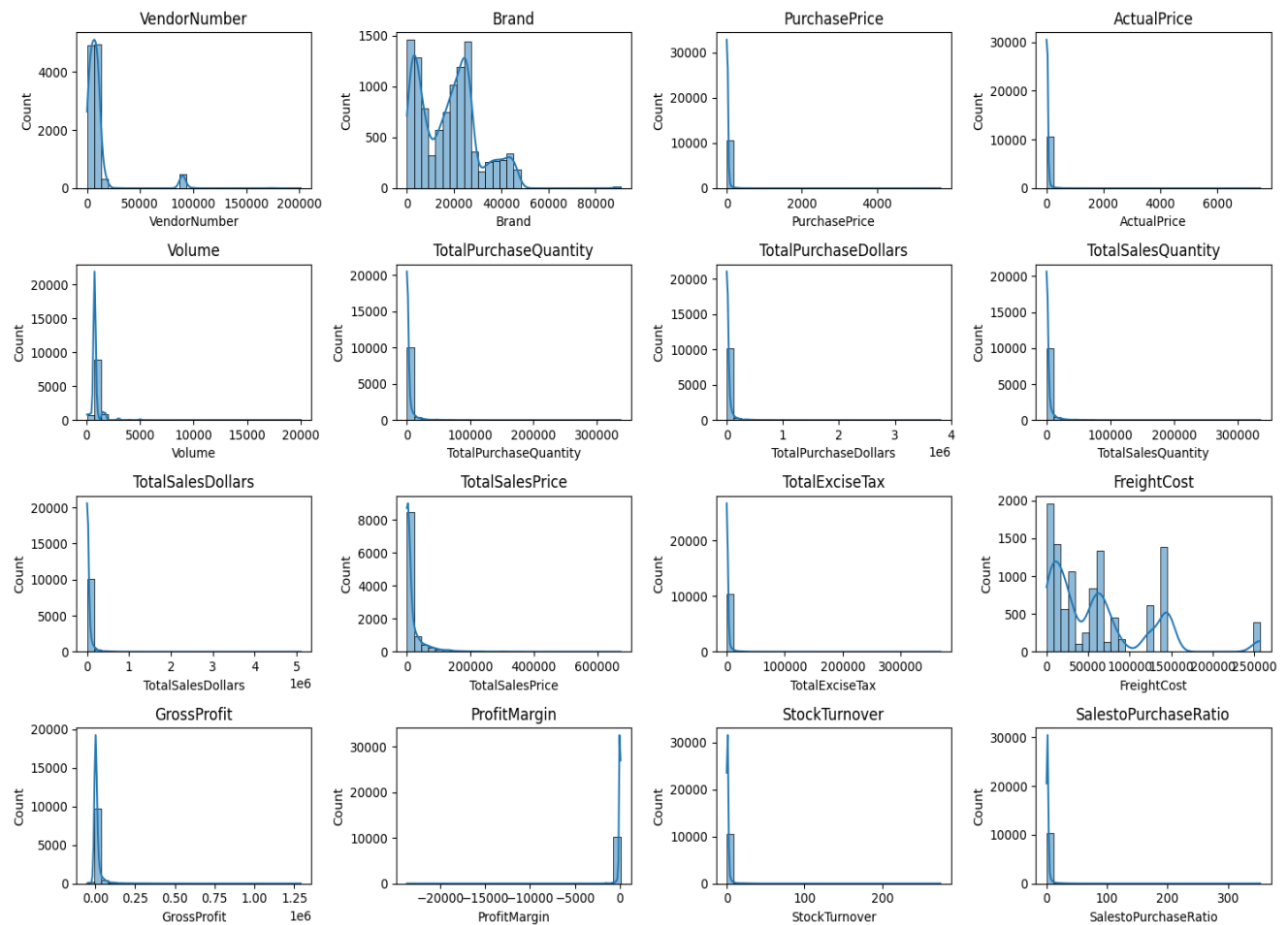


Exploratory Data Analysis Insights

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalestoPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02



Negative and Zero Values

The summary statistics revealed several important observations regarding negative and zero values within the dataset:

Gross Profit: The minimum gross profit value is -52,002.78, indicating that certain products or transactions were sold at a loss. This may occur when products are purchased at a higher cost or sold at discounted prices below their purchase cost.

Profit Margin: The minimum profit margin is negative infinity ($-\infty$), suggesting situations where sales revenue was either zero or lower than the associated costs. These cases require further investigation as they may indicate data quality issues or unprofitable transactions.

Total Sales Quantity and Total Sales Dollars: Both variables contain minimum values of 0, indicating that some products were purchased but never sold. Such products may represent slow-moving, obsolete, or excess inventory that ties up working capital.

Outliers Identified Through High Standard Deviation

Several variables exhibited substantial variation, indicating the presence of potential outliers and unusual business patterns:

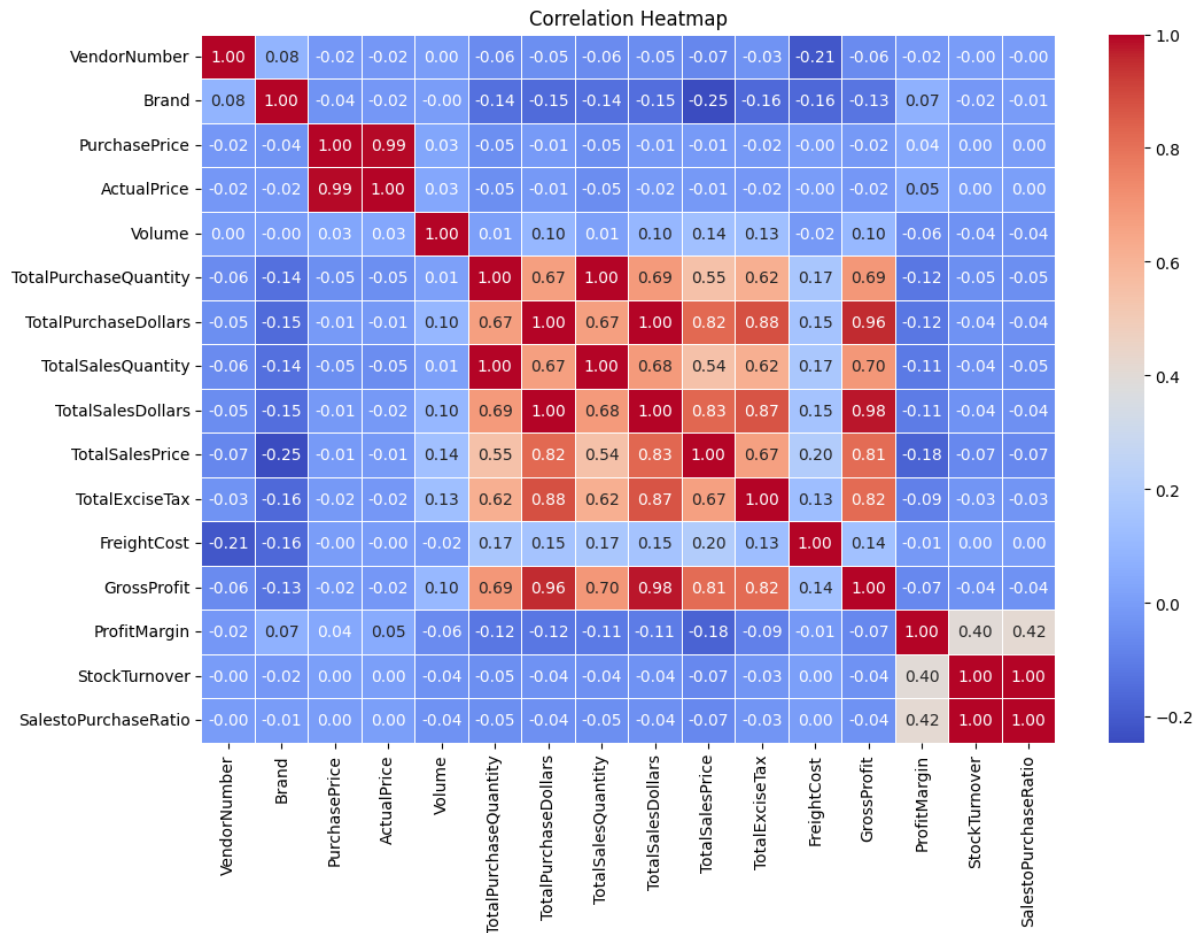
- **Purchase Price and Actual Price:** The maximum values (**5,681.81** and **7,499.99**, respectively) are significantly higher than their average values (**24.39** and **35.64**). This suggests the presence of premium or high-value products within the inventory portfolio.
- **Freight Cost:** Freight costs vary considerably, ranging from **0.09** to **257,032.07**. This large variation may indicate differences in shipment sizes, transportation methods, supplier locations, or logistics inefficiencies.
- **Stock Turnover:** Stock turnover ranges from **0** to **274.5**, demonstrating substantial differences in inventory movement across products. A turnover value greater than **1** indicates that sales quantities exceed current purchase quantities, potentially due to inventory being fulfilled from existing stock. Conversely, low turnover values may highlight products with slow sales velocity and inventory accumulation.

Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- **Gross Profit ≤ 0** (to exclude transactions leading to losses).
- **Profit Margin ≤ 0** (to ensure analysis focuses on profitable transactions).
- **Total Sales Quantity = 0** (to eliminate inventory that was never sold).

Correlation Insights



Purchase Price vs. Total Sales Dollars & Gross Profit: Weak correlation (-0.012 and -0.016), indicating that price variations do not significantly impact sales revenue or profit.

Total Purchase Quantity vs. Total Sales Quantity: Strong correlation (0.999), confirming efficient inventory turnover.

Profit Margin vs. Total Sales Price: Negative correlation (-0.179), suggesting increasing sales prices may lead to reduced margins, possibly due to competitive pricing pressures.

Stock Turnover vs. Gross Profit & Profit Margin: Weak negative correlation (-0.038 & -0.055), indicating that faster stock turnover does not necessarily equate to higher profitability.

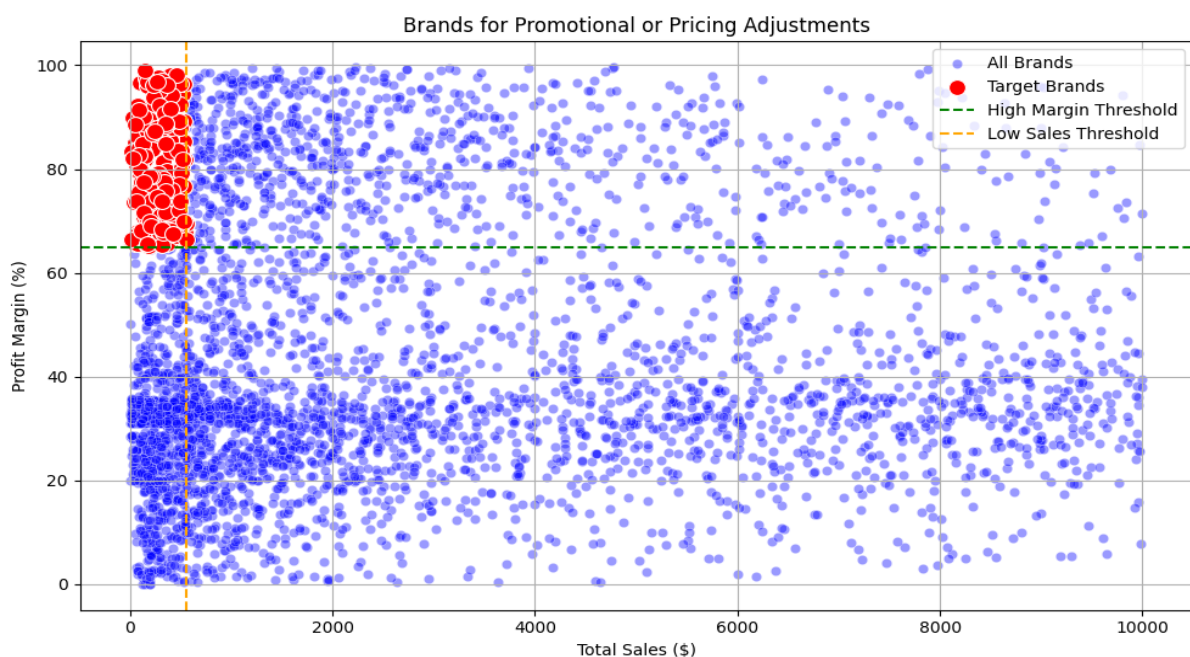
Research Questions & Key Findings

1.Brands for Promotional or Pricing Adjustments

Brand with low Sales but High Profit Margins:

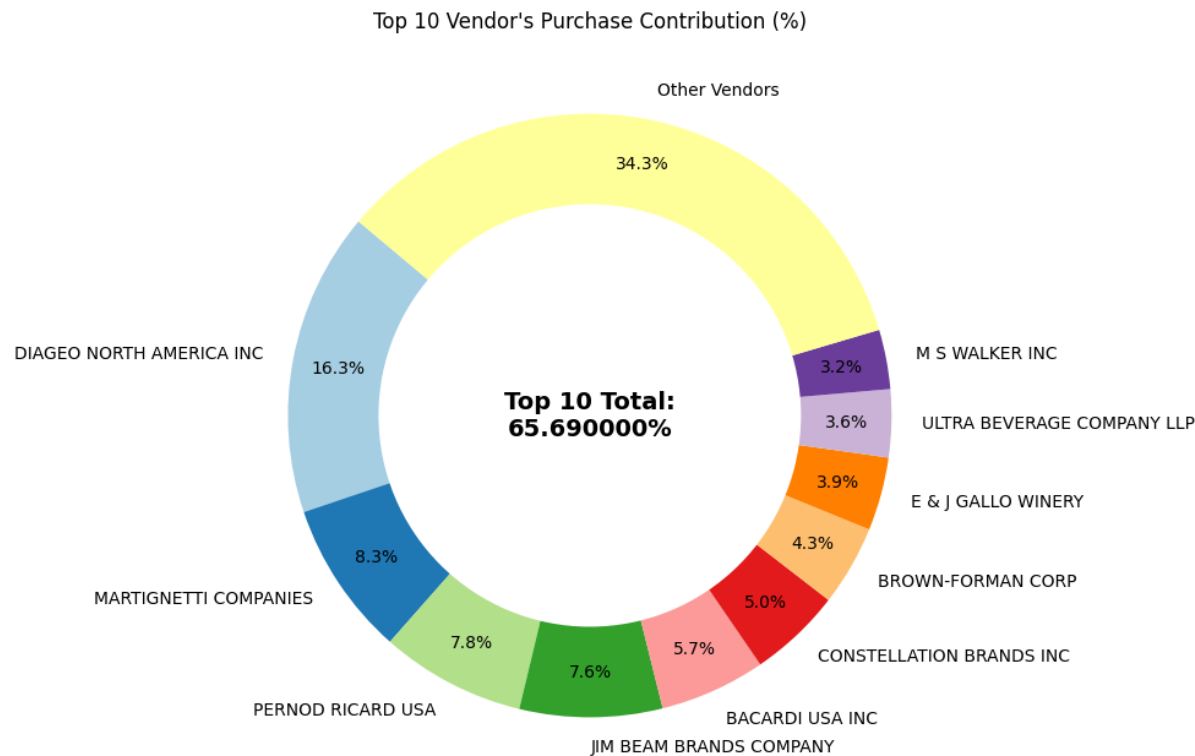
	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournel 10	558.87	69.434752

198 Brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.



2. Top Vendors by Sales & Purchase Contribution

The top 10 vendors contribute 65.69% of total purchases, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



3. Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a 72% lower unit cost (\$10.78 per unit vs. higher unit costs in smaller orders).

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

UnitPurchasePrice	
OrderSize	
Small	39.057543
Medium	15.486414
Large	10.777625

4. Identifying Vendors with Low Inventory Turnover

Total Unsold Inventory Capital: \$2.71M

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability.

Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

StockTurnover		VendorName UnsoldInventoryValue	
VendorName			
		25	DIAGEO NORTH AMERICA INC 722.21K
ALISA CARR BEVERAGES	0.615385	46	JIM BEAM BRANDS COMPANY 554.67K
HIGHLAND WINE MERCHANTS LLC	0.708333	68	PERNOD RICARD USA 470.63K
PARK STREET IMPORTS LLC	0.751306	116	WILLIAM GRANT & SONS INC 401.96K
Circa Wines	0.755676	30	E & J GALLO WINERY 228.28K
Dunn Wine Brokers	0.766022	79	SAZERAC CO INC 198.44K
CENTEUR IMPORTS LLC	0.773953	11	BROWN-FORMAN CORP 177.73K
SMOKY QUARTZ DISTILLERY LLC	0.783835	20	CONSTELLATION BRANDS INC 133.62K
TAMWORTH DISTILLING	0.797078	61	MOET HENNESSY USA INC 126.48K
THE IMPORTED GRAPE LLC	0.807569	77	REMY COINTREAU USA INC 118.60K
WALPOLE MTN VIEW WINERY	0.820548		

5. Profit Margin Comparison: High vs. Low-Performing

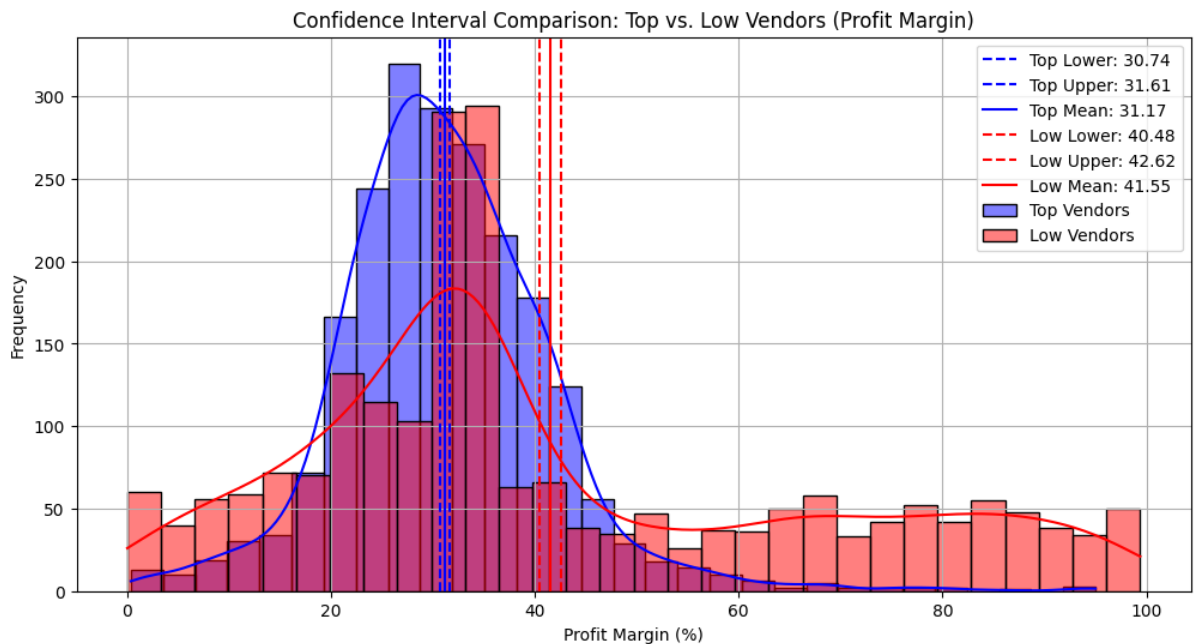
Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%), Mean: 31.17% Low Vendors' Profit Margin (95% CI): (40.48%, 42.62%), Mean: 41.55%

Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

- Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.

- Low-performing vendors: Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.



6. Statistical Validation of Profit Margin Differences

Hypothesis Testing:

- **H₀ (Null Hypothesis):** No significant difference in profit margins between top and low-performing vendors.
- **H₁ (Alternative Hypothesis):** A significant difference exists in profit margins between the two vendor groups.

Result: The null hypothesis is rejected, confirming that the two groups operate under distinctly different profitability models.

Implication: High-margin vendors may benefit from better pricing strategies, while top-selling vendors could focus on cost efficiency.

Final Recommendations

- Re-evaluate pricing for low-sales, high-margin brands to boost sales volume without sacrificing profitability.
- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.
- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales, or revising storage strategies.
- Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.

- By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.